

A Holistic Design Framework for a Modular Knee Exoskeleton System

Rajiv V. Joshi¹, Ilseung Park¹, Jimin An¹, Dawit Lee², and Inseung Kang¹

Abstract—Lower-limb robotic exoskeletons have advanced considerably through improved mechatronic designs and intelligent control architectures. However, most existing platforms, whether open-source or proprietary, depend on machined or custom-fabricated hardware that is expensive, slow to produce, and difficult to modify. These constraints make it hard for research groups to reproduce, adapt, or iterate on published designs, slowing progress on the controllers meant to drive them. This paper introduces PEAK, a fully 3D-printed, open-source knee exoskeleton built to be manufactured quickly and inexpensively using widely available desktop tools. The platform pairs a low-level embedded controller with a high-level computer, allowing new control strategies to be developed independently of hardware changes. To demonstrate the platform’s capabilities, we trained a knee-assistance controller entirely in a musculoskeletal simulation environment and deployed it on the physical device without collecting any real-world training data beforehand. Human subject testing during a squatting task showed that the resulting controller meaningfully reduced joint loading and muscular effort relative to squatting without assistance. This platform gives research groups a faster, more adaptable starting point for exoskeleton development.

Index Terms—Exoskeletons, wearable robotics, imitation learning, open-source, 3D printing.

I. INTRODUCTION

MOBILITY is fundamental to independence, yet millions of people worldwide struggle to walk, stand, or climb stairs due to age-associated musculoskeletal decline, neurological disease, injuries, or the physical demands of their occupation. Robotic exoskeletons, wearable devices that use actuators and sensors to assist human movement, have emerged as a promising intervention for addressing mobility-related challenges. The knee is a particularly important target joint, bearing the majority of the load during everyday activities such as standing up from a chair, navigating stairs, and squatting to lift heavy objects [1]–[4]. Exoskeleton hardware is typically built to validate a specific controller, and controller design has advanced quickly in recent years. Data-driven approaches now support increasingly sophisticated behaviors [5]–[9], and individual labs have developed capable custom hardware to demonstrate them [10]–[16]. This pairing of controller and purpose-built hardware, however, creates a gap for the broader research community. Cross-validating and benchmarking a new controller against existing approaches ideally requires access to the hardware those approaches were developed on. This access is rarely available: most exoskeleton platforms are

either not open-sourced or rely on complex, costly machined hardware that new labs cannot easily reproduce.

Open-source platforms have emerged in response to this limitation, aiming to give researchers validated baselines rather than requiring each lab to design a system from scratch. Williams et al. presented OpenExo [17], a versatile, modular exoskeleton platform that can serve as an assistive interface for the hip, ankle, or upper limbs. Nesler et al. introduced the Modular Backdrivable Lower-limb Unloading Exoskeleton (M-BLUE) [18], which applies backdrivable, compact quasi-direct-drive actuators to conventional hip and knee orthoses through simple aftermarket modifications, making the design replicable without custom fabrication for each joint configuration. Lee et al. released the CAD design and training data behind an AI-driven universal lower-limb exoskeleton controller [19], which uses a deep-learning-based gait phase and slope estimator to continuously modulate assistance across walking, ramps, and stairs without a separate mode classifier for each terrain. Together, these platforms mark significant progress toward making exoskeleton hardware and software accessible and standardized across research groups.

Despite this progress, these open-source platforms remain difficult to reproduce and iterate on quickly. OpenExo’s manufacturing pipeline spans thermoforming, machining, and 3D printing across a large number of tools. M-BLUE relies on custom-fabricated sheet-metal modifications and expensive Dephy FASTER drivers. Lee et al.’s platform adds a separate hip orthotic interface, additional machined components, and a backpack-mounted Jetson Nano for onboard compute. In each case, specialized fabrication capabilities, added cost, and assembly complexity limit how easily another lab can build, modify, or extend the design. Non-open-source knee exoskeletons show a related pattern from the opposite direction: strong performance with no path to reuse. Sarkisian et al. and Gunnell et al. reported EMG-controlled knee exoskeletons that meaningfully improved sit-to-stand transitions in stroke patients [20], [21], and Zhang et al. reported a single-source, dual-drive design using a clutch-and-Bowden-cable transmission that achieved effective assistance and metabolic reduction during locomotion [22]. All three rely on custom-machined or carbon-fiber components and tightly integrated electromechanical packaging, and none of these designs have been released. Across both open and closed platforms, the same pattern persists: substantial machining, specialized manufacturing steps, or complex electromechanical packaging make these devices costly, slow, and difficult for other groups to build, modify, or iterate on. When a design is not accessible at all, even strong performance results remain isolated to the group that produced

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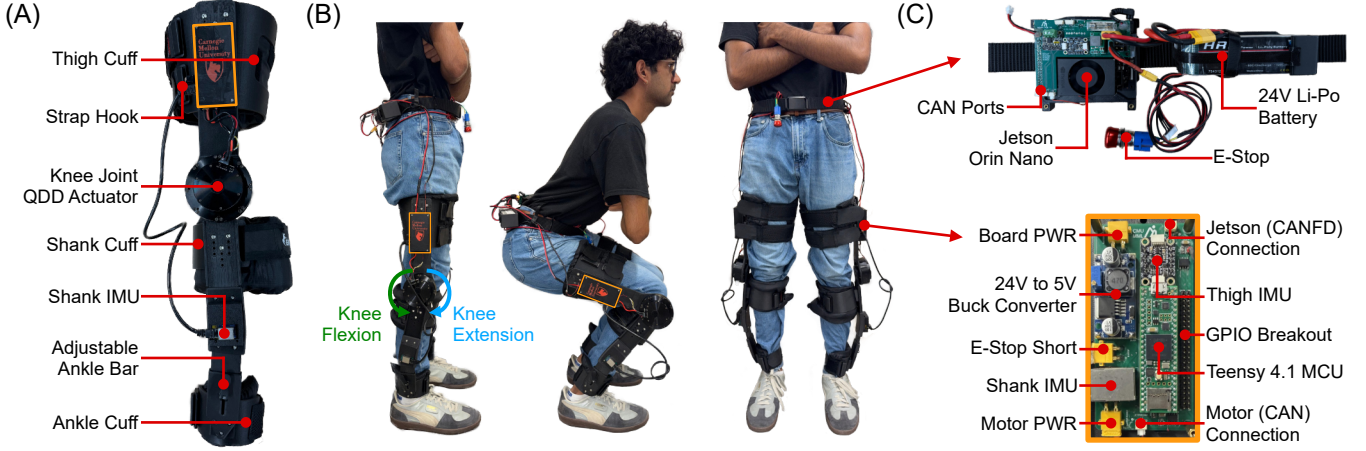


Fig. 1. **The PEAK exoskeleton hardware interface, fully 3D-printed using PAHT-CF filament with a three-cuff design optimized for extension-torque assistance.** (A) Sagittal-plane view of the unilateral device interface, with straps, hooks, and cuffs. (B) A participant wearing the device (photographed with consent), shown across three views, left to right: sagittal standing, sagittal squatting, and frontal standing. (C) The high-level control board, built around a Jetson Orin Nano, and the low-level control board below it, featuring the Teensy 4.1 microcontroller and exposed PCB, with embedded systems.

them.

In this work, we present a fully 3D-printed, open-source knee exoskeleton designed for low-cost, in-house fabrication and rapid iteration. We chose 3D printing over machined or aftermarket-modified construction specifically because it removes the specialized tooling, vendor lead times, and per-part cost that limited reproducibility in the platforms described above. Every structural component is printed from a small number of material spools and assembled with off-the-shelf fasteners, eliminating the need for machining, custom sheet-metal work, or other specialty components. Our embedded system architecture separates low-level sensing and motor control from high-level decision-making, allowing data-driven controllers to run on a dedicated high-level computer or, in principle, directly on the low-level microcontroller itself [23]–[25]. This separation allows new control strategies to be developed and tested independently of hardware changes.

We release the platform in full, including CAD files, PCB designs, and both low-level and high-level code, together with supporting documentation, so that other labs can adopt and extend the design without first reconstructing it from scratch. This approach builds on a broader trend toward experiment-free exoskeleton development, in which controllers are trained entirely in simulation and deployed without physical human trials [26]. To validate the platform without relying on extensive real-world data collection, we adopted a simulation-to-real pipeline: a musculoskeletal simulation generates reference squatting kinematics and corresponding knee joint moments, which are then distilled into a lightweight sensor-based estimator that runs directly on the exoskeleton’s onboard sensors. Together, these design choices allow the hardware, embedded software, and control policy to be reproduced, modified, and iterated on at a pace that existing machined and closed-source platforms cannot support.

For further validation, we tested the simulation-trained controller on the physical exoskeleton during squatting, comparing performance with and without the device across a group of participants. Using motion capture and surface EMG,

we found that the exoskeleton reduced peak knee extension moment relative to squatting without the device, with this reduction observed consistently across participants. Vastus medialis activation was similarly reduced when wearing the exoskeleton, while gastrocnemius activation showed no consistent change between conditions. Peak knee flexion angle was slightly reduced with the exoskeleton, consistent with the device sharing part of the mechanical demand of the squat. Together, these results indicate that a controller trained entirely in simulation, without any real-world data collection, transferred successfully to the physical platform and provided meaningful biomechanical assistance during a functional movement. Overall, the contributions of this work are: (i) an open-source, inexpensive, fully 3D-printed knee exoskeleton platform; (ii) a two-tier hierarchical control architecture; and (iii) experimental validation of a simulation-trained controller through EMG and motion-capture measurements showing meaningful biomechanical effects, reduced muscular effort and joint loading during squatting.

II. EXOSKELETON INTERFACE

A. Device Interface

PEAK (Powered Exoskeleton for Assisting Knee) is an open-source and fully 3D printed knee exoskeleton platform for bilateral or unilateral knee torque assistance in biomechanical research and assistive locomotion applications. The following sections describe the device implementation for hardware, low-level control, and the high-level controller.

B. Mechanical Design and Actuation

The PEAK mechanical interface consists of three cuffs spanning the thigh, shank, and ankle segments (Fig. 1), fabricated by fused deposition modeling (FDM) using PAHT-CF (Bambu Lab, Shenzhen, China), a carbon fiber reinforced polyamide filament. Print orientation and infill were selected to exploit

the mechanical efficiency of shell-dominant structures under bending loads. For a beam in bending, stress is governed by

$$\sigma = \frac{My}{I}, \quad (1)$$

where σ is the bending stress, M is the applied bending moment, y is the distance from the neutral axis, and I is the second moment of area of the cross-section, given by

$$I = \int y^2 dA. \quad (2)$$

Because I scales with the square of distance from the neutral axis, material positioned far from the neutral axis, such as the outer walls of a printed part, contributes disproportionately more to bending resistance than material near the core. Each cuff was therefore printed with solid outer walls and minimal infill, concentrating material where it most effectively resists bending, analogous to the flange-dominant design of a structural I-beam [27]. Each cuff follows a conical curvature matched to the geometry of the lower limbs and is longitudinally adjustable to accommodate variation in leg lengths. The ankle cuff serves as the primary anchor point, constraining distal migration of the device during dynamic tasks. Attachment to the user is achieved with a custom strap, cut to size from off-the-shelf webbing, that interfaces with integrated tabs on the cuff structure. The padding on the inner surface of each cuff is oversized relative to the strap width to eliminate chafing of the strap against the skin. Each cuff is lined with a custom multilayer padding, consisting of an outer 3D mesh fabric layer for breathability and soft skin contact. Inside the padding, an inner polyurethane upholstery foam layer evenly distributes applied force and eliminates pinching of the cuff against the user during assistance. The unilateral device mass, excluding the waist-mounted electronics, totals approximately 1.2 kg.

Torque is generated by a quasi-direct drive (QDD) actuator (AK80-9 V3.0; CubeMars, Nanchang, Jiangxi, China), which integrates a brushless DC motor, a 9:1 planetary gear reduction, a magnetic encoder, and a field oriented control (FOC) driver into a single unit mounted directly at the knee joint. The actuator has a rated torque of 9 Nm, a manufacturer-specified peak torque of 22 Nm, a rated speed of 390 rpm, a mass of 490 g, and a minimal backdrive torque of 0.51 Nm. However, the actuator's stock driver imposes an internal current limit that caps usable output at approximately 18 Nm; achieving the full manufacturer-rated torque would require a higher-current driver board or direct support from the manufacturer. These parameters make the actuator well suited for exoskeleton applications, a suitability confirmed by its use in numerous previous exoskeleton studies [18], [19], [28]. The motor output shaft couples to a rigid shank bar, and the motor body is fixed to a thigh bar. Extension torque is transmitted to the joint as tangential shear forces applied through these bars via the cuff and strap interface. A comparison against existing knee exoskeleton platforms is provided in Table I.

C. Embedded System and Control

Onboard control is divided across two computing tiers (Fig. 2). A Teensy 4.1 microcontroller (PJRC, Sherwood, OR,

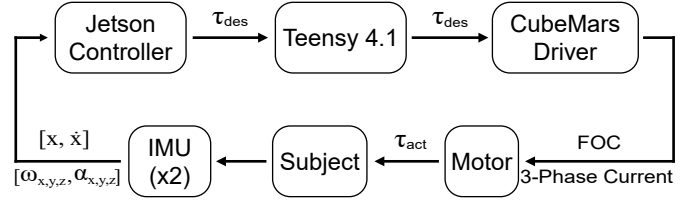


Fig. 2. **General controller framework of the high-level and low-level control loops.** The low-level Teensy control loop runs at 1000 Hz, recording the latest encoder position and velocity along with thigh and shank IMU accelerations ($\alpha_{x,y,z}$) and velocities ($\omega_{x,y,z}$). The high-level Jetson control loop runs at 100 Hz, requesting the current device state at the start of each loop and outputting the desired torque at the end. All communications between the two control loops are interrupt-driven.

USA), mounted locally on each leg module, executes low-level control at 1000 Hz. At each control cycle, the Teensy polls two ICM-20948 inertial measurement units (IMUs; TDK InvenSense, San Jose, CA, USA), sagittally on the thigh and shank, via SPI to acquire linear acceleration and angular velocity for state estimation. Although the IMUs natively sample at approximately 1200 Hz, readings are polled at the 1000 Hz control loop rate. The Teensy also communicates with the AK80-9 actuator over CAN bus, issuing torque commands to the actuator's onboard FOC driver and receiving motor state feedback (position, velocity, and torque) for logging.

A Jetson Orin Nano (NVIDIA, Santa Clara, CA, USA), mounted on the waist belt, executes high-level control at 100 Hz, matching the sampling rate of the treadmill, motion capture, and electromyography (EMG) instrumentation. Synchronization across all systems was achieved using a GPIO cable connected to the Jetson, with two GPIO trigger pulses marking the start and end of each trial for offline data alignment. At the start of each control cycle, the Jetson polls both Teensy microcontrollers over a shared CANFD bus to acquire the latest IMU and motor state from each leg module, passes the relevant signals to the active controller (Section II-D), and transmits the resulting torque command back to each Teensy. End-to-end latency across the full bilateral CANFD network is less than 80 ms. A watchdog timer on each Teensy acts as a safety mechanism: if a valid torque command is not received from the Jetson within the expected cycle window, the motor is commanded to 0 Nm. This returns the device to a passive, backdrivable state in the event of any communication failure. During experimental trials, sensor and control data were logged onboard the Jetson and organized by trial, then transferred to an external computer via SSH for offline analysis.

Motor and computing power are supplied by a 6 cell, 24 V, 3300 mAh lithium polymer battery (HRB; Shenzhen, China) mounted on the waist belt, providing approximately 79.2 Wh of capacity. Based on the estimated average power draw of the onboard computing and actuation hardware, this corresponds to an estimated continuous operating duration of approximately 2 hours. The total mass of the bilateral system, including the waist belt, Jetson, and battery, is approximately 3.4 kg.

TABLE I
COMPARISON WITH EXISTING KNEE EXOSKELETON PLATFORMS

Device	Bilateral Mass (kg)	Peak Torque (Nm)	VasMed Reduction (%)	Tasks Tested	Manufacturing Method	Actuation Method	Open Sourced
This Study	3.4	18	26.7	SQ	FDM	QDD	Yes
Nesler et al. [18]	3.3	25	*	LL	Modified Orthosis	QDD	Yes
Lee et al. [19], [29]	4.4	15	19.3	LG, RA, RD, SA, SD, STS	Machined / Orthosis	QDD	Yes
Zhang et al. [22]	3.2	18	—	LG, RA, SA	Machined	Clutch, Bowden Cable	No
Boon et al. [30]	2.6	55	32	SA	Machined / CF	Ball Screw Drive	No
Chen et al., Yu et al. [11], [12]	3.5	54	39.4–43.2	KNL, SQ	Machined/FDM	QDD	No

*Nesler et al. reported a 78.9% reduction, but from a single subject lifting a 12.5 kg weight; not directly comparable.

LG: level-ground walking; **RA:** ramp ascent; **RD:** ramp descent; **SA:** stair ascent; **SD:** stair descent; **SQ:** squat; **STS:** sit-to-stand; **LL:** lifting/lowering; **KNL:** kneeling.

D. Simulation-based Knee Joint Moment Estimator

The controller was developed through a two-stage learning framework that connects reference human kinematics to sensor-based knee assistance. First, a DeepMimic-style [31] learning policy was trained to reproduce reference squatting kinematics in a physics-based musculoskeletal simulation. The resulting muscle-driven motion provided synchronized kinematics and knee joint moments for training a moment estimator. Second, these simulation-generated data were distilled into a temporal convolutional network (TCN) [32] that estimates knee moment from the recent history of knee angle and thigh and shank angular velocities. During deployment, the estimator receives measurements from the exoskeleton’s onboard sensors, and its output is scaled to generate assistive knee torque.

E. Musculoskeletal Simulation and Policy Training

Reference squatting kinematics were obtained from the open-source dataset [33]. The kinematics data were low-pass filtered using a sixth-order Butterworth filter with a cutoff frequency of 4 Hz, applied bidirectionally. Joint velocities were computed by differentiating the filtered joint coordinates.

A musculoskeletal simulation environment was implemented in SCONE [34] using the Hyfydy physics engine [35]. The sagittal-plane model comprised seven degrees of freedom, nine muscles in a single lower limb, and a lumbar actuator. A reinforcement learning policy was trained using soft actor-critic (SAC) [36] to reproduce the reference squatting motion. The policy received simulated joint and muscle states, contact information, and future reference kinematics. Its actions specified muscle excitations and a lumbar target angle, which was tracked by a proportional–derivative controller.

The training objective combined motion imitation with penalties on muscular effort and abrupt changes in actuator inputs:

$$r_t = w_p r_{p,t} + w_v r_{v,t} + w_e r_{e,t} - w_a c_{a,t} - w_u c_{u,t}, \quad (3)$$

where $r_{p,t}$, $r_{v,t}$, and $r_{e,t}$ reward agreement with the reference joint coordinates, joint velocities, and foot position relative to the pelvis, respectively. These tracking rewards were formulated as exponential functions of the corresponding squared errors. The effort cost $c_{a,t}$ was defined as the mean squared

muscle activation, and $c_{u,t}$ penalized changes in muscle excitations and normalized lumbar torque between consecutive steps. The weights were set to $w_p = 0.60$, $w_v = 0.15$, $w_e = 0.25$, and $w_a = w_u = 2.0$.

The policy operated at 100 Hz, with episodes initialized from sampled reference states and limited to 10 s. Episodes were terminated early when the pelvis deviated excessively from the reference trajectory. The resulting policy generated muscle-driven squatting motion from which knee muscle moments were obtained for subsequent distillation.

F. Distillation

The simulation-trained policy relied on information that is unavailable from the wearable sensors, including muscle states, contact forces, and future reference motion. To obtain a deployable moment estimator, a TCN was trained using sensor-equivalent kinematics and knee muscle moments collected from deterministic rollouts of the trained policy.

At each simulation step, the sagittal-plane angular velocities of the thigh and shank and the knee joint angle were recorded as

$$\mathbf{x}_t = [\omega_{\text{thigh},z,t} \quad \omega_{\text{shank},z,t} \quad q_{\text{knee},t}]^T. \quad (4)$$

These signals correspond to measurements available from the thigh and shank IMUs and the knee actuator encoder. The synchronized knee muscle moment was extracted from the musculoskeletal model and scaled by a fixed normalization factor to form the regression target y_t . Each training example consisted of a 95-sample input window ending at time t and the moment target at that same time:

$$\hat{y}_t = f_\theta(\mathbf{x}_{t-94}, \dots, \mathbf{x}_t). \quad (5)$$

The TCN used residual blocks with dilated causal convolutions to extract temporal features from the input history, followed by a linear readout that produced a scalar moment estimate. Input channels were standardized using stored means and standard deviations. Gaussian noise was added to the gyroscope channels, and knee-angle offsets were introduced during training to expose the estimator to sensor noise and calibration variation.

The student network was trained by minimizing the mean squared error between its predictions and the simulation-derived moment targets:

$$\mathcal{L}_{\text{distill}} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2. \quad (6)$$

Training used the AdamW optimizer with a learning rate of 5×10^{-4} and a batch size of 64 for 50 epochs. The checkpoint with the lowest validation loss was selected. This distillation procedure transferred the relationship between squatting kinematics and simulated knee muscle moment into a sensor-based estimator requiring only the current and preceding wearable-sensor measurements.

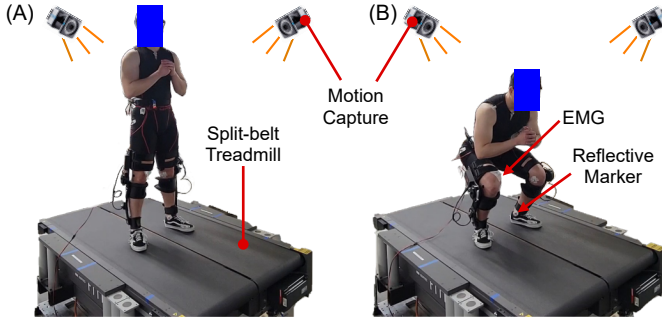


Fig. 3. **Laboratory setup used for the human experimental validation.** (A) A participant wearing the exoskeleton in a standing posture, with the instrumented treadmill and its embedded force plates for measuring ground reaction forces. (B) The same participant performing an assisted squat, with example placements of EMG electrodes and reflective motion-capture markers.

III. HUMAN EXPERIMENTAL VALIDATION

A. Experiment Protocol

Seven healthy participants (all males; age: 26.29 ± 2.6 years; body mass: 76.5 ± 8.7 kg; height: 1.77 ± 0.04 m) were recruited under a protocol approved by the Carnegie Mellon University Institutional Review Board (Protocol STUDY2026_00000339). All participants provided written informed consent before the experiment.

Data were collected in a marker-based motion capture laboratory equipped with a 10-camera system (Vero v2.2; Vicon, Oxford, UK) and a split belt instrumented treadmill with two embedded force plates (FIT5; Bertec, Columbus, OH, USA) (Fig. 3). A total of 24 markers were placed at anatomical landmarks and joint centers for full-body tracking and subsequent inverse kinematics processing in OpenSim [37]. Two surface EMG sensors, recorded using a 4-channel Trigno Quattro Research+ system (Delsys, Natick, MA, USA), were placed over the vastus medialis (VasMed), a primary contributor to knee extension torque during squatting, and the medial gastrocnemius (GasMed). Marker trajectories and ground reaction forces were sampled at 100 Hz.

A static calibration trial (standing upright, feet shoulder width apart, arms at the sides, palms facing anteriorly) was then recorded to generate a subject-specific musculoskeletal model. Following a 5-minute rest, the participant completed a

60 s unassisted (NoExo) squatting trial with one foot centered on each force plate. Squatting was paced to a 40 BPM metronome, with each tick corresponding to a change in squat direction. The final 10 s of each trial were retained for analysis to exclude the transient adaptation period at trial onset.

The participant then rested for 5 minutes while the bilateral exoskeleton was donned and anatomical joint alignment was verified. With the knees in full extension, the motor encoder zero reference was established. After 5 to 8 familiarization squats and a further 5-minute rest, the assisted squatting trial was performed under a protocol identical to the unassisted condition.

B. Data Processing and Statistical Analysis

Kinematic and kinetic data were processed using OpenSim [37] with participant-scaled versions of the modified RajagopalLaiUhrich2023 musculoskeletal model [38]. The NoExo trials were analyzed using the corresponding NoExo model, whereas the Exo On trials were analyzed using a model augmented to represent the mass and inertia of the exoskeleton. A total exoskeleton mass of 3.40 kg was distributed across the lower limbs of the model (0.80 kg on the pelvis, 0.91 kg on each thigh, and 0.39 kg on each shank), and each segment's diagonal moments of inertia were scaled in proportion to its increased mass. Inverse dynamics was subsequently performed using the corresponding condition-specific model. Both inverse kinematics and dynamics were low-pass filtered at 6 Hz. The exoskeleton torque command was synchronized with the motion-capture data and subtracted from the inverse-dynamics knee moment for the Exo On condition. Individual squat cycles were identified from the vertical pelvis trajectory after fourth-order, zero-phase Butterworth low-pass filtering at 6 Hz. The final 10 complete squat cycles from each condition were time-normalized and averaged within each participant.

VasMed and GasMed EMG signals were resampled to 1000 Hz and demeaned before being processed using a fourth-order, zero-phase Butterworth band-pass filter between 10 and 499 Hz. The filtered signals were full-wave rectified, and linear envelopes were obtained using a fourth-order, zero-phase Butterworth low-pass filter at 8 Hz. Maximum voluntary contraction (MVC) recordings were processed using the same filtering procedure. To limit the influence of transient artifacts in the MVC trials, values within the upper 1% of the rectified signal were replaced using neighboring signal values before envelope generation. The MVC reference for each muscle was defined as the maximum value of a 250-ms moving average of the processed MVC envelope. Experimental EMG signals were then normalized to the corresponding reference and expressed as a percentage of MVC (%MVC). The EMG signals were segmented using the squat-cycle boundaries obtained from the kinematic data, time-normalized, and averaged across the final 10 cycles for each participant and condition. Peak VasMed and GasMed activation values were defined as the maximum values of each participant's resulting mean squat-cycle profile.

To test our exoskeleton performance, we performed two-sided paired *t*-tests on the corresponding peak values of knee joint angle, joint moment, and EMG signals. Participant-level

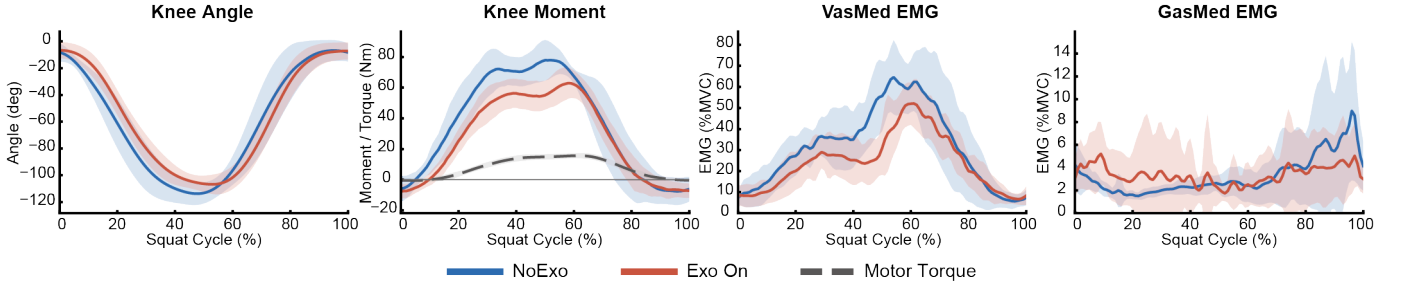


Fig. 4. **Group-averaged knee kinematics, kinetics, and muscle activation during squatting.** Panels show knee angle, knee extension moment with commanded motor torque, vastus medialis (VasMed) activation, and medial gastrocnemius (GasMed) activation over the normalized squat cycle. Solid curves represent the group means across participants ($n = 7$), and shaded regions represent ± 1 standard deviation. Blue and orange indicate the NoExo and Exo On conditions, respectively, and the dashed gray curve indicates commanded motor torque during the Exo On condition. EMG amplitudes are expressed as a percentage of each participant's maximum voluntary contraction (%MVC).

peak values were obtained from each participant's mean profile across 10 squat cycles. Data are reported as mean $\pm$ standard deviation, with statistical significance defined as $p < 0.05$.

IV. RESULTS

A. Knee Joint Kinematics & Dynamics

We compared the knee joint kinematics and dynamics between the Exo On and NoExo conditions over the normalized squat cycle. The overall knee angle profiles followed a similar flexion pattern in both conditions (Fig. 4). However, peak knee flexion was lower in the Exo On condition than in the NoExo condition, decreasing from $114.5 \pm 7.7^\circ$ to $107.1 \pm 7.4^\circ$, respectively ($p = 0.019$).

Across participants, the average knee moment profiles also demonstrated a lower magnitude during the Exo On condition than that of the NoExo condition (Fig. 4). Peak knee extension moment decreased significantly from 81.9 ± 10.1 Nm in the NoExo condition to 64.3 ± 9.5 Nm in the Exo On condition, corresponding to a 21.5% reduction ($p = 0.006$; Fig. 5). The reduction in peak extension moment was observed in all seven participants. During the Exo On condition, peak commanded motor torque was 15.7 ± 1.7 Nm and occurred at $60.1 \pm 5.6\%$ of the normalized squat cycle (Fig. 4).

B. Muscle Activation

We further compared the muscle activation profiles of VasMed and GasMed between conditions. The group-averaged VasMed activation profile was lower during the Exo On condition than during the NoExo condition (Fig. 4). Peak VasMed activation decreased from 77.1 ± 17.6 %MVC to 56.5 ± 12.3 %MVC, corresponding to a 26.7% reduction ($p = 0.016$; Fig. 5). Participant-level peak VasMed activation decreased in six of the seven participants. In contrast, the GasMed profiles exhibited greater between-participant variability (Fig. 4). Peak GasMed activation was 9.86 ± 6.40 %MVC in the NoExo condition and 8.29 ± 4.85 %MVC in the Exo On condition (Fig. 5). No statistically significant between-condition difference was detected ($p > 0.05$).

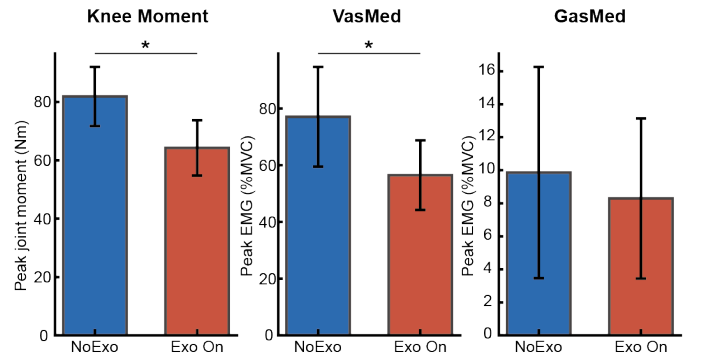


Fig. 5. **Comparison of peak knee moment and muscle activation between conditions.** Bars represent the group means of peak knee extension moment, VasMed activation, and GasMed activation in the NoExo and Exo On conditions. Error bars represent ± 1 standard deviation across participants ($n = 7$). Asterisks indicate a significant between-condition difference ($p < 0.05$, paired t -test).

V. DISCUSSION

This study introduced PEAK, an open-source, inexpensive, 3D printable knee exoskeleton platform, and demonstrated its application to assisted body weight squatting in a controlled motion capture environment. The following summary analyzes the proposed advantages and limitations of the device.

The performance of the AK80-9 V3.0 actuator confirms that it is a well-suited, off-the-shelf actuator for exoskeleton applications. As a quasi-direct-drive actuator capable of 18 Nm peak torque and requiring only 0.51 Nm to back-drive, it provides sufficient torque output while remaining highly backdrivable. Because the knee experiences substantially greater extension than flexion moments during squatting tasks, PEAK's cuff configuration prioritized extension torque transfer. Early design iterations revealed cuffs spanning only the sagittal regions of the leg were insufficient for transferring actuator torque to the limbs. Under higher torque loads, medial and lateral rotation of the device around the leg propagated a compounding error in the data-driven controller through incorrect sensor readings. Wrapping the cuffs around both front and back of the leg mitigated this rotation by converting a portion of the applied shear force into normal force against the limb. The resulting cuff configuration, shown in Fig. 1A,

reliably transferred the full 18 Nm peak torque to the user with minimal device migration and accurate sensor peripheral readings. Given that no material deformation was observed across all seven participants' trials, PAHT-CF appears to be a suitable material for device durability, withstanding the high temperatures generated by the actuator during extended usage. Future experimentation with higher-torque actuators will confirm whether the current design paradigm is sufficient beyond 18 Nm of applied torque.

The choice to place a dedicated Teensy microcontroller on each leg module, rather than a single shared low-level controller, may appear overspecified relative to typical exoskeleton architectures, which commonly use two microcontrollers or a single microcontroller paired with one high-level computer. This choice was intentional. By keeping each leg's low-level control self-contained, the platform supports future development of controllers that run entirely onboard the microcontroller, without requiring the Jetson, in either a bilateral or unilateral configuration. This flexibility directly supports the microcontroller-deployable control strategies discussed in the introduction, in which a sufficiently compressed model could eventually replace the current Jetson-hosted estimator for lower-power or lower-cost applications.

As shown in Table I, PEAK's peak torque and bilateral mass are comparable to entry-level performance among existing open-source knee exoskeleton platforms. Notably, this is achieved despite PEAK being the only fully 3D-printed design in the comparison. PEAK's 26.7% reduction in peak quadriceps activation is achieved using a comparatively modest 18 Nm of peak torque. This result falls within the range reported by other knee exoskeletons validated with EMG. It is comparable to Boon et al.'s 32% reduction at 55 Nm during stair ascent and Yu et al.'s 39.4–43.2% reduction at 54 Nm during squatting, though below Nesler et al.'s 78.9% reduction at 25 Nm. PEAK's reduction also exceeds Lee et al.'s 19.3% at a comparable 15 Nm. Because PEAK's applied torque is substantially lower than each of these comparison platforms, it remains unclear whether a similarly scaled torque budget would yield a comparable or greater reduction in muscle activation. Evaluating PEAK's performance at higher torque levels is therefore an important direction for future work.

The controller's main limitation is its narrow task scope. The moment estimator was trained only on squatting kinematics from a single reference dataset, and its performance on other tasks remains untested. The TCN maps a short history of knee angle and thigh and shank angular velocity to a single knee moment target, learned specifically from squat-pattern simulation rollouts. This mapping reflects the joint coordination and effort penalties used to train the underlying imitation policy, which were tuned to reproduce squatting kinematics rather than general lower-limb movement. As a result, the estimator's output is only meaningful for motions with similar joint coordination patterns to squatting, and may not generalize to tasks with different knee-hip-ankle coupling, such as walking or stair negotiation. This is not a fundamental limit of the framework, however. The same simulation-to-real distillation pipeline could be retrained on kinematic data from other tasks, extending the estimator beyond squatting without

changing the underlying architecture.

A second limitation emerged during early deployment. Initial lateral rotation of the device around the limb introduced noise into the thigh and shank IMU readings. This noise propagated into the estimator's input and degraded the accuracy of the predicted knee moment, reducing the effectiveness of the commanded torque. Because the estimator relies entirely on real-time IMU and encoder signals, its output is only as reliable as these measurements. Once the cuff redesign addressed this rotation, IMU readings stabilized and torque output improved substantially. This highlights a tight coupling between mechanical design and controller performance in this platform. Future work will expand the simulation framework to additional tasks, such as sit-to-stand, walking, and stair negotiation, to evaluate the estimator's generalizability across a broader range of movements.

The reduction in peak knee extension moment indicates that the exoskeleton meaningfully supplemented the internal joint demand during squatting. This finding is corroborated by the concurrent reduction in VasMed activation, since the quadriceps femoris group is the primary contributor to knee extension moment during this task. The consistency of both reductions across nearly all participants suggests the assistance was not limited to a subset of favorable cases, but reflects a general offloading effect. In contrast, GasMed activation showed no significant between-condition difference. This is expected, as the ankle joint was not directly assisted, and the GasMed contributes comparatively little to knee extension torque during squatting. The modest but significant decrease in peak knee flexion under assistance may reflect a small compensatory adjustment in squat depth or strategy, though this effect was not directly assessed and warrants further investigation. Together, these results suggest that the PEAK exoskeleton reduced the neuromuscular burden of the targeted muscle group without inducing compensatory activation in an unassisted, adjacent muscle. These findings should be interpreted with the study's sample size of seven participants in mind, which limits the strength of any broader generalization to other populations or task variations. Additionally, this validation was limited to healthy young adults, and the device's effectiveness for the older or clinical populations motivating this work remains untested. Extending validation to these populations is an important direction for future work.

VI. CONCLUSION

Fast, low-cost fabrication is key to making exoskeleton hardware reproducible across research groups. This work contributes a fully 3D-printed knee exoskeleton, an open two-tier control architecture, and a controller trained entirely in simulation. Together, these enable PEAK, an open-source platform that can be built and modified far faster than existing machined or closed-source designs. Deployed on the physical device without any real-world training data, the controller reduced peak knee extension moment and quadriceps activation during assisted squatting. Future work should address the limitations discussed above and evaluate the platform across a broader set of tasks and participants.

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